

Representing inputs as vectors

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

ML Builder Expo

Without opening the old lesson, explain one key idea from ML Builder Expo. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Convert before encoding for Representing inputs as vectors. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Representing inputs as vectors. Proposed explanation: “Order does not matter”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new schema independently

A new toy battery task uses [temperature in degrees C, voltage in V, age in days]. Record K has voltage 5, age 12, temperature 18. Record L has temperature 20, voltage 4, age 0. Encode both; state the vector dimension and table shape. Age zero is valid. The future failure label is also supplied in the data file: should it be appended?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Encode the records

Write A, B and C as ordered pairs from the lesson. State the dimension of each and the shape of the stacked table.

My reasoning and evidence

6. W2 Convert before encoding

A new record has width 60 mm and height 10 mm. Give its vector in the agreed cm schema, and name one error in $[60, 10]$.

My reasoning and evidence

7. W3 Handle missing information

Width is 4 cm and height is missing. A teammate writes $[4, 0]$. Explain why this is invalid and give the action required by this schema.

My reasoning and evidence

8. W4 Apply a new schema independently

A new toy battery task uses [temperature in degrees C, voltage in V, age in days]. Record K has voltage 5, age 12, temperature 18. Record L has temperature 20, voltage 4, age 0. Encode both; state the vector dimension and table shape. Age zero is valid. The future failure label is also supplied in the data file: should it be appended?

My reasoning and evidence

9. W5 Explain a collision

Two parcels have the same width and height but different material and target class. Explain why adding more examples alone cannot let this deterministic two-feature model tell these two inputs apart.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled representing inputs as vectors evidence sheet to Image-pattern Network Investigation. Show how you can encode a record using a fixed feature order and units. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Representing inputs as vectors

Encode the same pixel measurements in a fixed vector order.

Toy image features are [left_brightness, right_brightness], each = 0..10. ImageA has left2, right7; imageB has left7, right2.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/1

Representing inputs as vectors

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Declare coordinate order

Coordinate0 is left brightness; coordinate1 is right brightness.



2. Encode both images

A becomes [2, 7]; B becomes [7, 2]. Their sums match but their spatial patterns differ.



3. Check input contract

Use the same order when training and predicting; record missing measurements rather than guessing.

CHANGE ONE THING

Swap only A's vector order before prediction.

Explain the changed result using the matching lesson.

Why does equal total brightness not make A and B identical inputs? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

13. Project portfolio milestone

Project: Image-pattern Network Investigation

Current milestone: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Encode the same pixel measurements in a fixed vector order.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Use the same order when training and predicting; record missing measurements rather than guessing.

1. Declare coordinate order
2. Check input contract
3. Encode both images

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Check input contract Use the same order when training and predicting; record missing measurements rather than guessing. A vector is an ordered

list, so the position of a number carries meaning. Here the first number always means left brightness. Adding the two measurements keeps total brightness but discards which side is brighter. A network receives the representation we give it; it cannot recover a lost distinction just because it has more layers.

Practice 2: explain the deciding evidence

Changed-case practice: Swap only A's vector order before prediction. Which conclusion is supported by the supplied facts?

1. The encoded input becomes [7, 2], numerically identical to B's original input.
2. We can decide the result without examining the changed input or the procedure.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The encoded input becomes [7, 2], numerically identical to B's original input. Changing coordinate order changes feature meaning even if the numbers are unchanged.

Repair a missing building block

Review ML Builder Expo (ai-4/36). Goal: Defend the project using traceable evidence. Short practice: Which three supporting items should be visible at the expo? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback

and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
